

TIM 172B, Lecture #3 (1/13/26)

1. SC Strategy / Design of the SC Network
 - Responsiveness / Efficiency
 - Implied Demand Uncertainty (IDU)
 - Process for developing SC strategy
2. HW #1 was discussed in L # 2
 - some additional suggestions (related to the SC Network: Prob 2d)
3. FAST diagram for SCM
4. On Thursday: assign seating for the class
5. We will require & take attendance periodically (for the TIM Minors)

1. SC strategy

(a) Performance : Responsiveness / Efficiency

There are 2 key performance attributes (or metrics) for the SC Network

(i) SC efficiency : cost of making, storing, and delivering the product to the customer.

Higher efficiency $\Leftrightarrow$ lower the costs

(ii) SC responsiveness : ability of the SC to respond rapidly to the following demands :

- Large range of products (product variety)
- Large changes in the quantity demand
- shorter lead times
- higher service levels
- highly innovative products

We capture the trade-off between eff. & resp. using a Resp./Eff. spectrum.

Resp. / Eff. spectrum

Eff. ←

→ Resp.

Highly eff. SC:
integrated steel mills
(e.g. mold for rocket engines):
extremely narrowly defined product for which production can be scheduled well in advance

somewhat eff. SC
Apparel manufactures
(Hanes, Jockey, ...)
production lead-times of several months

somewhat responsive SC:
Automobiles
Lead times are several weeks to months

Highly responsive
DELL:
(Build-to-order customized PCs in a few days.

(b) Implied Demand Uncertainty (IDU)

IDU is different from Demand Uncertainty (DU)

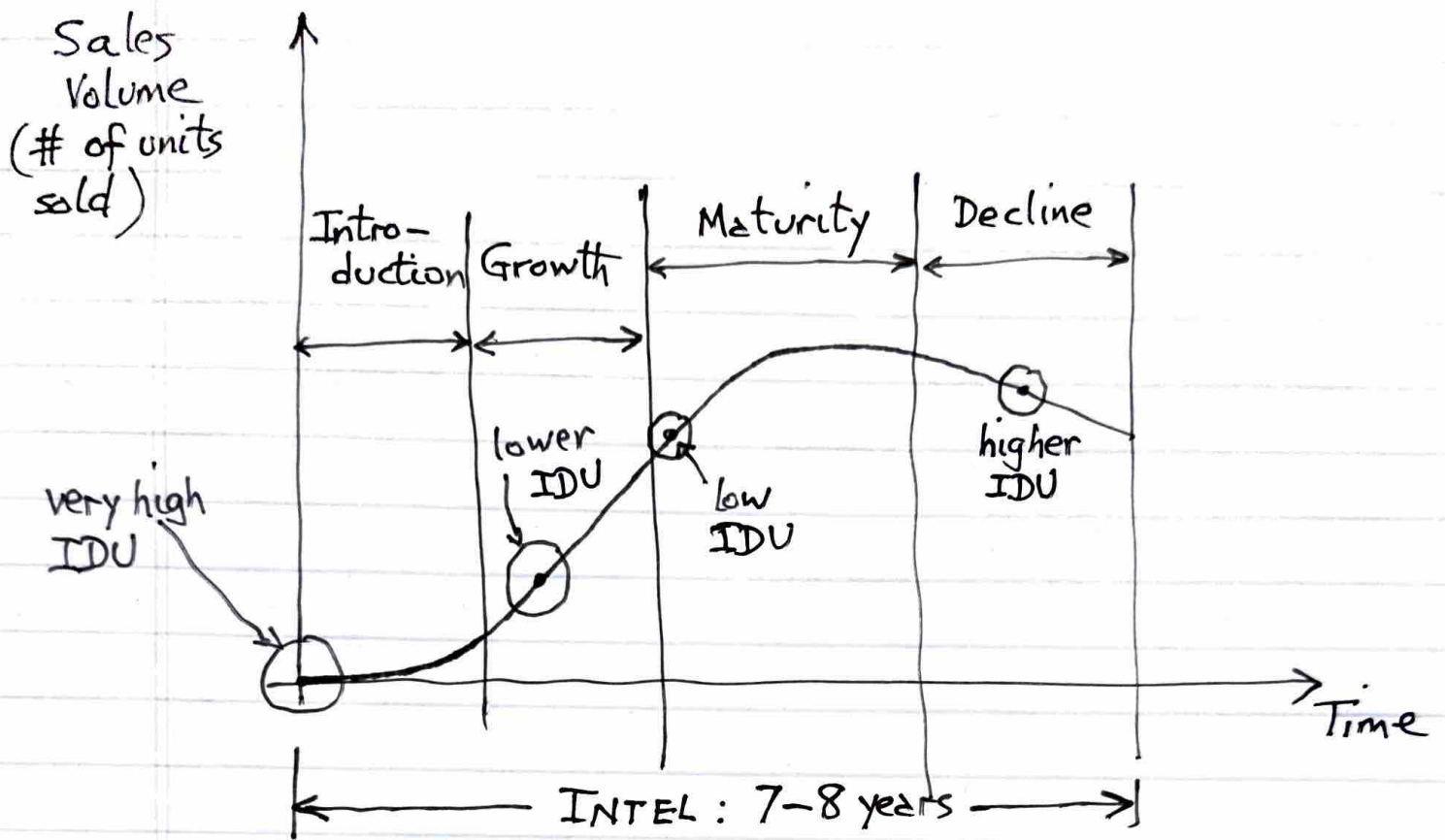
(i) demand uncertainty : uncertainties in the nominal
(DU) quantity of the product demanded by the customer

example :

annual demand for product X = $\underbrace{100,000 \text{ units}}_{\substack{\text{nominal} \\ \text{or} \\ \text{average} \\ \text{demand}}} \pm \underbrace{20\%}_{\text{DU}}$

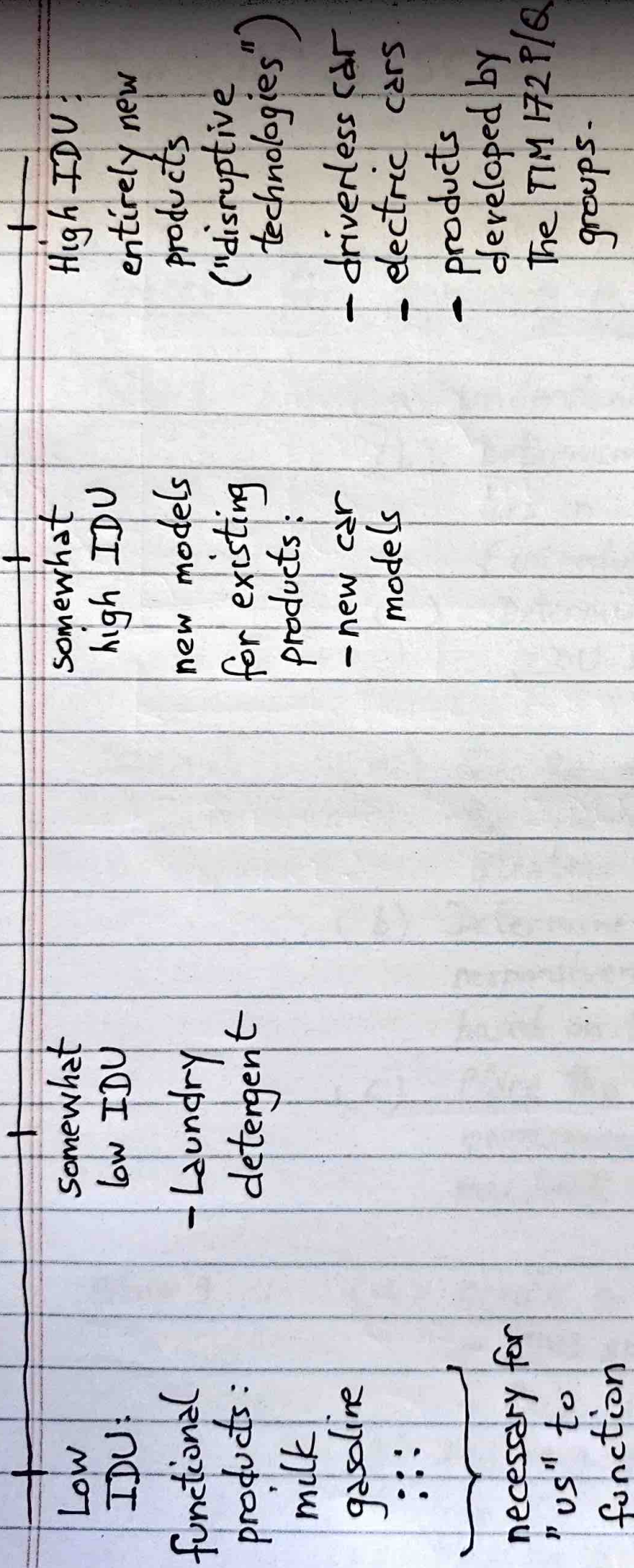
(ii) Implied Demand Uncertainty (IDU)

refers to the uncertainty in demand for the product implied by the customer need for the product during the life-cycle of the product in its market.



→ Determine the IDU spectrum for the product.

IDU spectrum



(c) Developing a SC strategy (aka achieving a zone of strategic fit in a SC)

Process for developing a SC strategy: 3 steps

Step 1 :

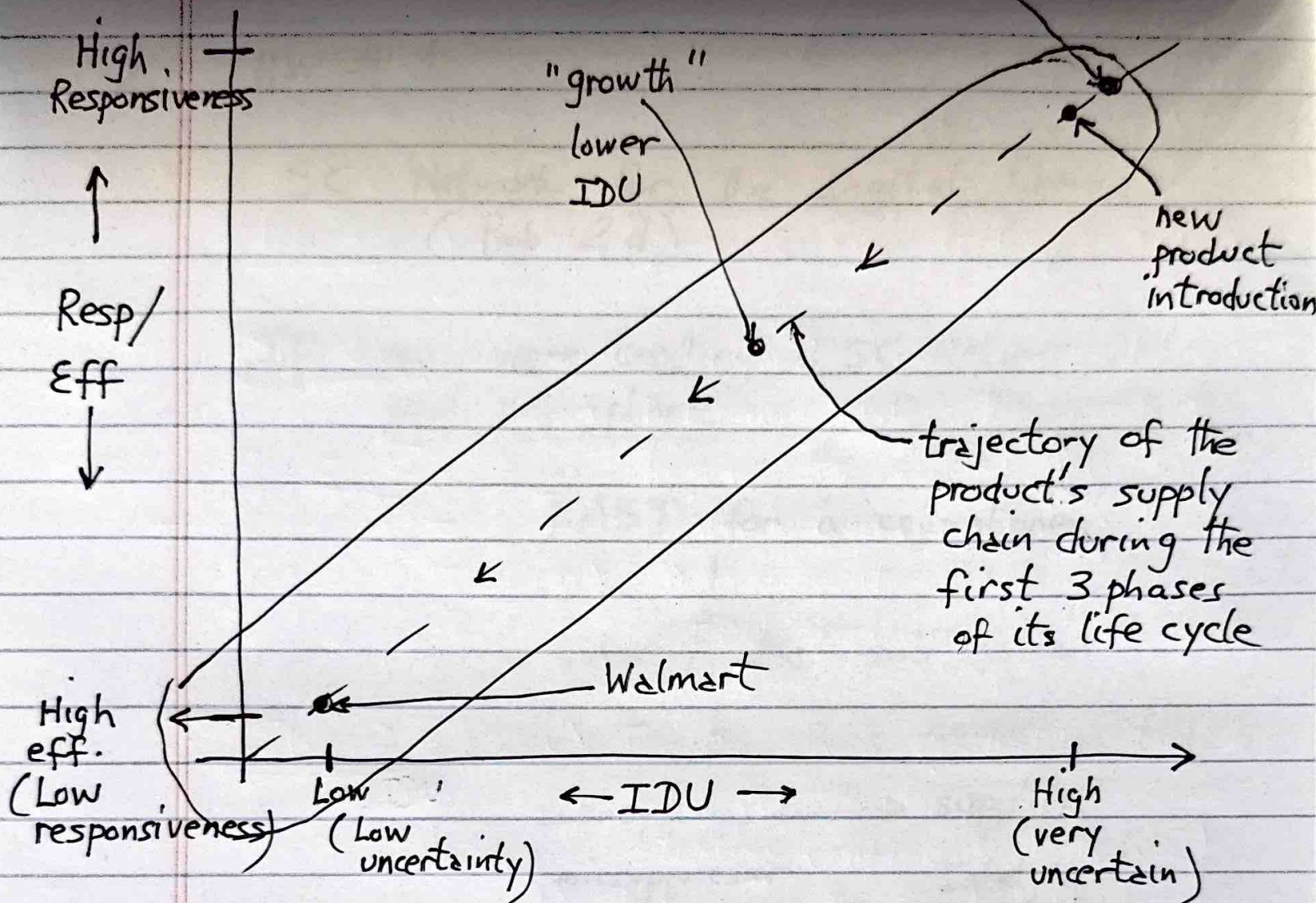
- (a) understand customer needs
- (b) Determine where the product lies in its market life-cycle (introduction, growth,)
- (c) Determine its position on its IDU spectrum

Step 2 :

- (a) For the given product, establish the corresponding competitive strategy (differentiated,)
- (b) Determine the trade-off between responsiveness & efficiency based on the competitive strategy
- (c) Place the product at the corresponding position on the resp./eff. spectrum

Step 3 :

- (a) create a 2-D space as follows
 - IDU spectrum on the X-axis
 - Resp./Eff. spectrum on the Y-axis
- (b) Define a "zone of strategic fit" in the 2-D space
- (c) Map the SC strategy for the product inside the "zone of strategic fit."



Comment : Companies like Dell, Walmart, Amazon, and Apple are both very efficient and highly responsiveness

- Information systems/technology can enable the company to both responsive & very efficient.

2. HW # 1

SC Network for The Digital Camera (Prob. 2d)

If you were creating a SC Network for your cell-phone

FAST for a cell-phone

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Identify the sub-systems

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sub-systems	↔	supplier
microprocessor		Intel,
digital camera		DC manufacturer
storage		Seagate, ...

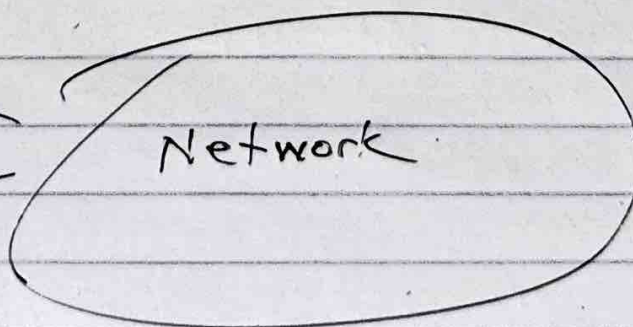
↓
SC Network

suppliers/
sub-systems

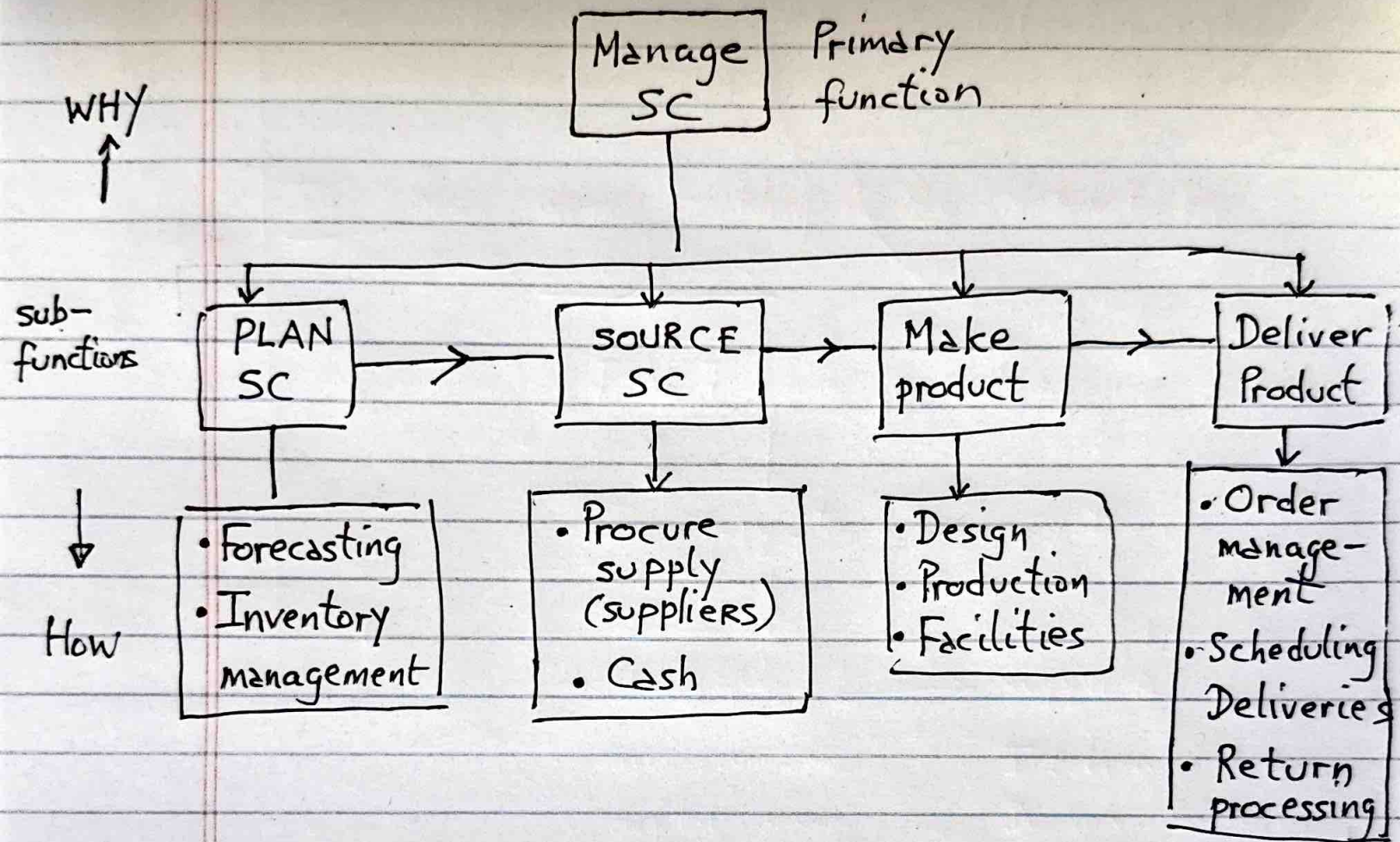
INTEL
[Microprocessor]

⋮

SEAGATE
[Storage]



3. pseudo-FAST representation of SCM



" Plan - Source - Make - Deliver " model.